

2025 年《信息安全技术》期末考查说明

考查方式：自行组队 1-3 人为一组完成，按所选题目要求提交相应报告、代码和视频。报告中注明每个组员的具体工作贡献。

提交截止时间：**7 月 13 日 23:59 分。**

提交方式：将课程报告、代码（包含 README）一起打包压缩后，命名为“姓名 1、姓名 2+学号 1+学号 2+信息安全技术期末考察”（比如“丁一、丁二+21307000、21307001+信息安全技术期末考察”）后，并发送至对应邮箱。**具体需要发至的邮箱将在分组完成之后公布。**

评分方案：根据提交的报告和代码综合评价，主要考虑：方案完整，步骤清晰，实验对比或创新改进，分析合理，总结全面。

期末考查题目： 课程报告

本课程的期末考核方式采用课程报告。具体要求如下：

1. 报告的内容：以下内容二选一即可。

(1) 选择一篇与课程所讲内容相关（例如密码学，数字水印，隐私推理等）领域的 CCF-B 以上的会议/期刊论文，并展开阅读，对该论文方案进行改进创新，并且通过实验验证改进效果，撰写并提交课程报告。

(2) 选择两篇与课程所讲内容相关（例如密码学，数字水印，隐私推理等），且在相同领域的信息安全领域的 CCF-B 以上的会议/期刊论文，并展开阅读，深刻理解并复现论文，对比这两篇论文的优缺点，撰写并提交课程报告。

2. 报告语言：中文。

3. 篇幅（采用中文，五号字，单倍行距）：

单独写作课程报告：不少于 5 页，不少于 5000 字。

4. 格式：课程报告应包含但不限于以下部分

(1) 背景和动机介绍，简要介绍研究的背景和动机；

(2) 两篇复现的方案或复现的方案及改进，详细叙述报告中采用的方案，并且介绍这两篇复现的方案之间的关系或改进方案的设计和原理；

(3) 实验验证和结果分析，基于你的代码对两个方案在标准评价指标下进行对比实验，并且给出你的实验分析；

(4) 代码运行效果的截图。

(5) 总结，对本次课程报告进行总结，给出自己的观点；

5. 注意：课程报告与实验应该是组内独立完成的工作，禁止抄袭。

部分可以参考的文献

数字水印技术

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- [2] Kirchenbauer J, Geiping J, Wen Y, et al. A watermark for large language models[C]//International Conference on Machine Learning. PMLR, 2023: 17061-17084.
- [3] Huo M, Somayajula S A, Liang Y, et al. Token-Specific Watermarking with Enhanced Detectability and Semantic Coherence for Large Language Models[C]//International Conference on Machine Learning. PMLR, 2024.
- [4] Yang Z, Zeng K, Chen K, et al. Gaussian Shading: Provable Performance-Lossless Image Watermarking for Diffusion Models[C]//IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024: 12162-12171.
- [5] Wen Y, Kirchenbauer J, Geiping J, et al. Tree-Ring Watermarks: Fingerprints for Diffusion Images that are Invisible and Robust[C]//Advances in Neural Information Processing Systems (NeurIPS), 2023.
- [6] Fernandez P, Couairon G, Jégou H, et al. The Stable Signature: Rooting Watermarks in Latent Diffusion Models[C]//IEEE/CVF International Conference on Computer Vision (ICCV), 2023: 22466-22477.
- [7] Zhang X, Li R, Yu J, et al. EditGuard: Versatile Image Watermarking for Tamper Localization and Copyright Protection[C]//IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024: 11964-11974.
- [8] Jia Z, Fang H, Zhang W. MBRS: Enhancing Robustness of DNN-based Watermarking by Mini-Batch of Real and Simulated JPEG Compression[C]//Proceedings of the 29th ACM International Conference on Multimedia (ACM MM), 2021: 41-49.
- [9] Zhang Y, Ni J, Su W, Liao X. A Novel Deep Video Watermarking Framework with Enhanced Robustness to H.264/AVC Compression (REVMARK)[C]//Proceedings of the 31st ACM International Conference on Multimedia (ACM MM), 2023: 8095-8104.
- [10] San Roman R, Fernandez P, Elsharh H, et al. Proactive Detection of Voice Cloning with Localized Watermarking (AudioSeal)[C]//International Conference on Machine Learning (ICML), 2024.

隐私推理技术

- [11] Kim D, Guyot C. Optimized privacy-preserving cnn inference with fully homomorphic encryption[J]. IEEE Transactions on Information Forensics and Security, 2023, 18: 2175-2187.
- [12] Peng H, Huang S, Zhou T, et al. Autorep: Automatic relu replacement for fast private network inference[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023: 5178-5188.
- [13] G. He, Y. Ren, G. He, G. Feng and X. Zhang, "PPNNI: Privacy-Preserving Neural Network Inference against Adversarial Example Attack," in IEEE Transactions on Services Computing, doi: 10.1109/TSC.2024.3399648.

- [14] Pang, Q., Zhu, J., Möllering, H., Zheng, W., & Schneider, T. (2024, May). Bolt: Privacy-preserving, accurate and efficient inference for transformers. In 2024 IEEE Symposium on Security and Privacy (SP) (pp. 4753-4771). IEEE.
- [15] Xu, T., Wu, L., Wang, R., & Li, M. PrivCirNet: Efficient Private Inference via Block Circulant Transformation. In The Thirty-eighth Annual Conference on Neural Information Processing Systems. NIPS 2024.
- [16] Nam, Kevin, et al. "{LOHEN}: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy." 34th USENIX Security Symposium (USENIX Security 25). 2025.
- [17] jie Lu, Wen, et al. "BumbleBee: Secure two-party inference framework for large transformers." NDSS 2025. The Internet Society, 2025.
- [18] Yang, Zhaomin, et al. "RBOOT: Accelerating Homomorphic Neural Network Inference by Fusing ReLU within Bootstrapping." IACR Cryptol. ePrint Arch. 2025 (2025): 1534.